Computer Vision 1

Introduction

Lecturer: Yen-Chia Hsu

Sep 2026

Credit: This slide is inherited from Dr. Martin Oswald and Dr. Dimitris Tzionas

University of Amsterdam

Today's Agenda

Part 1:

- What is computer vision?
- Challenges
- Brief History
- Applications

Part 2:

- Course Structure
- Q&A

What is Computer Vision?

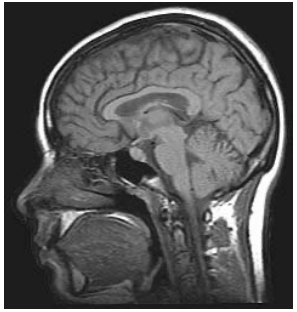
What is Computer Vision?

Computer Vision concerns building **artificial** vision systems that obtain useful information from **image data**.

The image data can take many forms, e.g., color images, depth images, video sequences, multiple views from multiple cameras, multi-dimensional medical scans

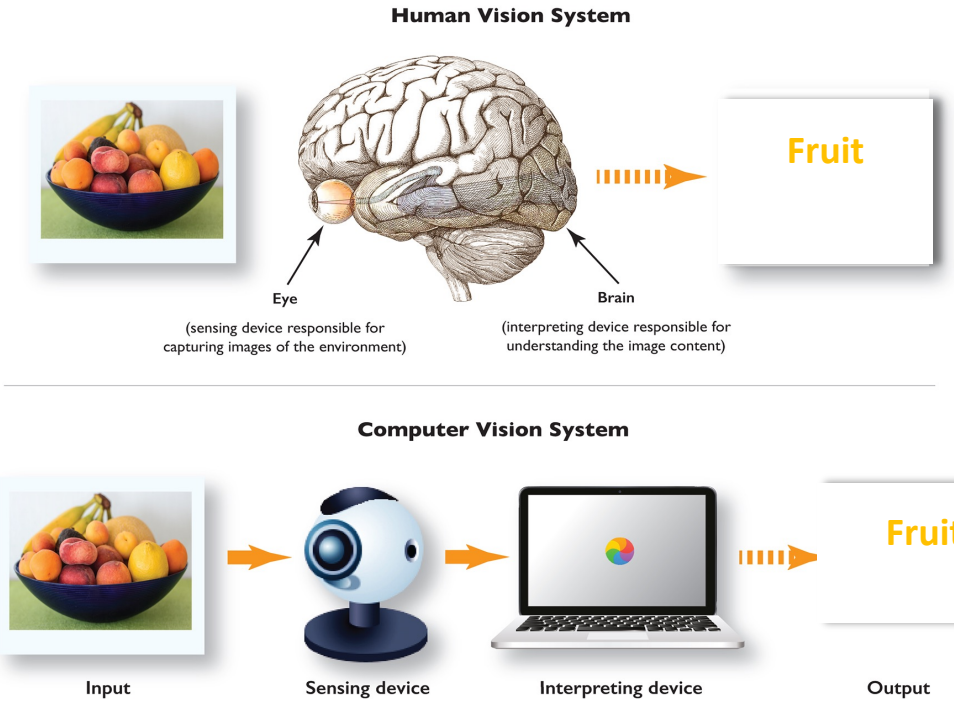






Computer Vision

- Enable computers to ‘see’ & ‘understand’ images
- For humans & animals this seems ‘effortless’
- But for computers this has proven to be challenging



Mohamed Elgendy, Deep Learning for Vision Systems, 2020

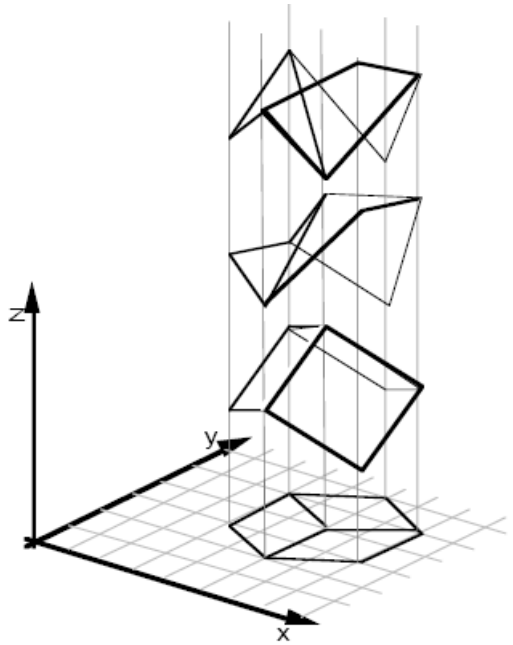
Challenges

Challenges – Projection Ambiguities

Ill-posed problem

Many 3D shapes project on the same 2D shape

We cannot recover 3D from 2D shapes (we miss 1D)



[Sinha and Adelson, 1993]

Challenges – Scale Ambiguities



https://richardwiseman.com

Challenges – Single Viewpoint





https://x.com/CSProfKGD/status/1724847975191523701

Challenges – Occlusion



Magritte
1957

slide by Fei Fei,
Fergus & Torralba

Challenges – Viewpoint Variations



Slide by Fei Fei, Fergus & Torralba

Michelangelo 1475-1564

Challenges – Illumination Conditions



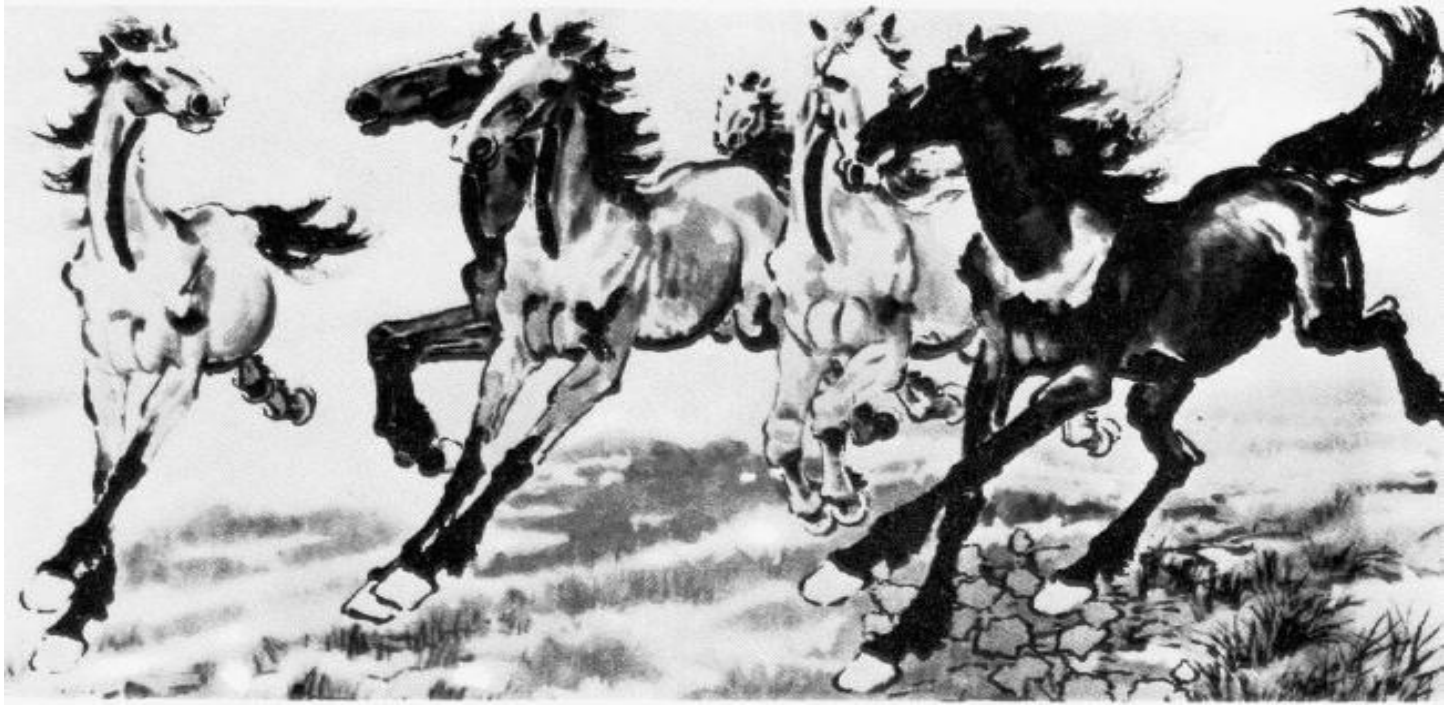


Slide credit:
S. Ullman

Challenges – Appearance Similarity & 1 View



Challenges – Deformable Shapes



Xu, Beihong 1943

Challenges – Background Clutter



Klimt
1913

slide by Fei Fei,
Fergus & Torralba

Challenges – Intra-class Shape Variance

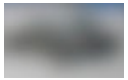


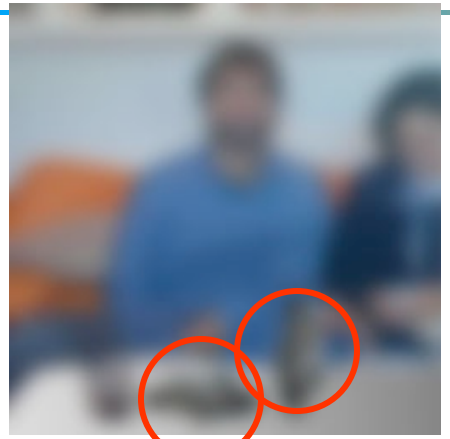


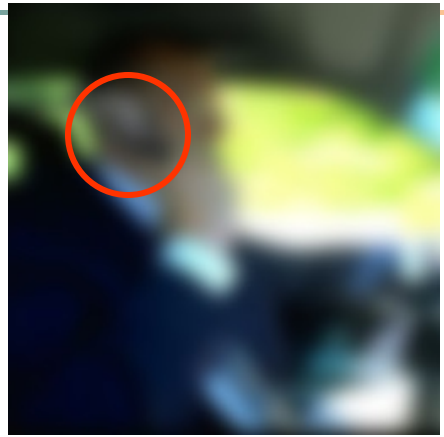


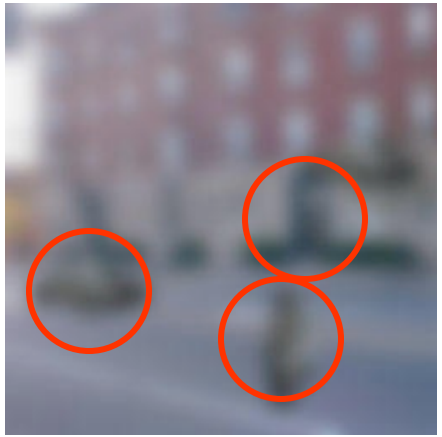

slide by Fei Fei, Fergus & Torralba

Challenges – Local Similarities











slide by Fei Fei,
Fergus & Torralba

Challenges – Depth Estimation



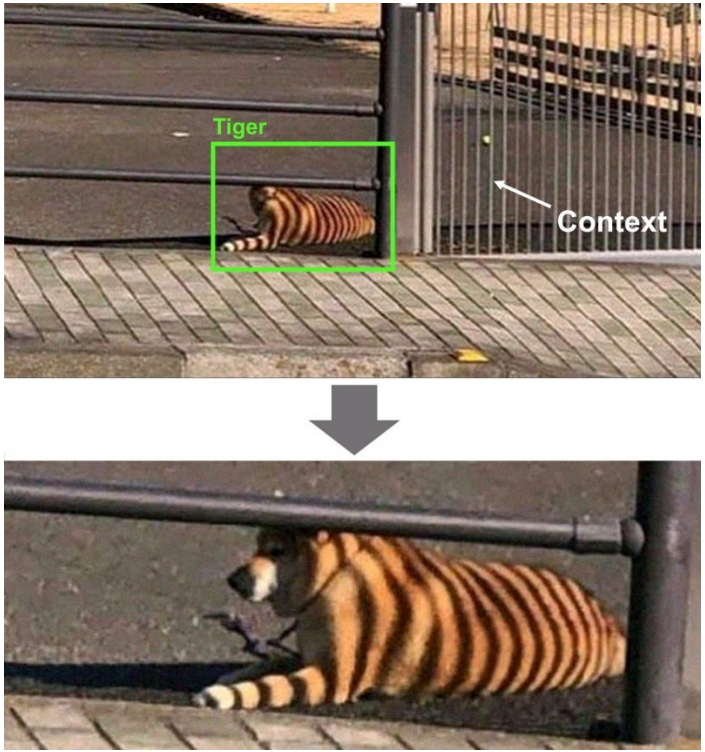
Slide Credit:
Alyosha Efros

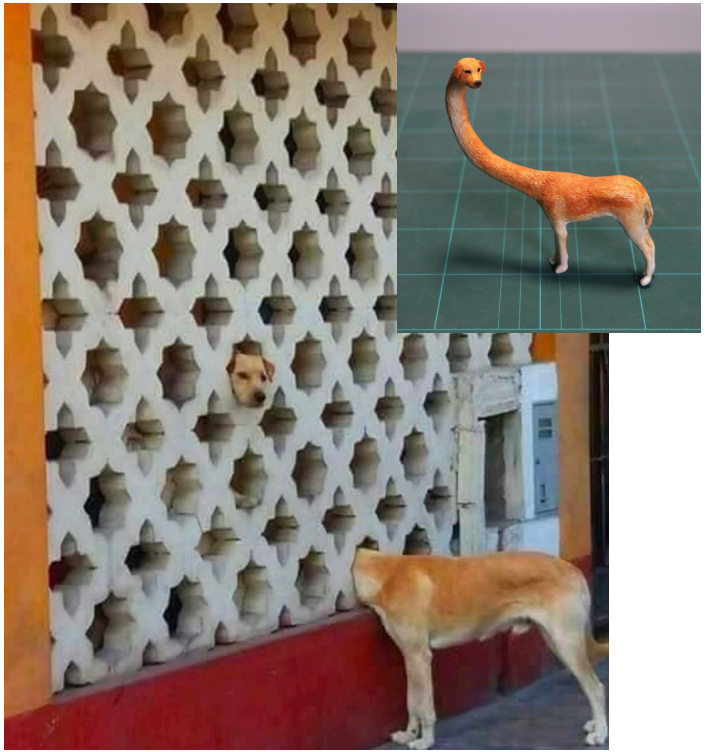
Challenges – Depth Estimation



[Marc Rober, YouTube]

Challenges in Scene Understanding

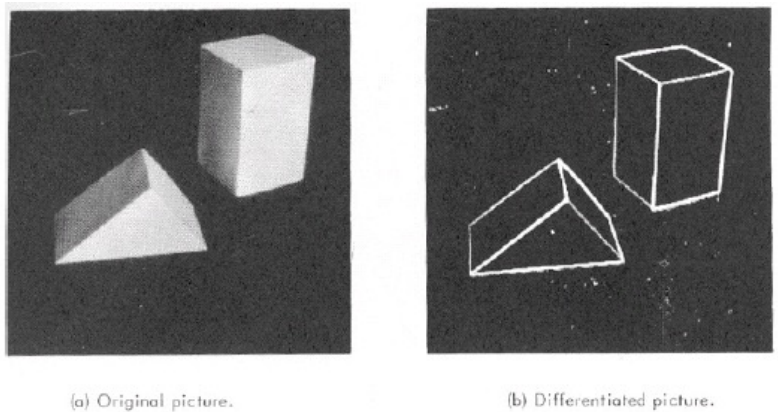




How to learn 3D **spatial context**
& **disentangle** various scene **factors**?

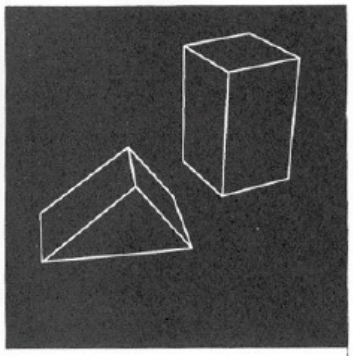
Brief History

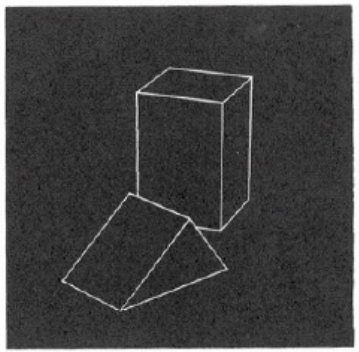
Origins



(a) Original picture.

(b) Differentiated picture.





L.G. Roberts,

'*Machine Perception of Three Dimensional Solids*'

PhD thesis

MIT, 1963

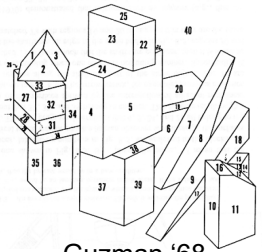
An (overly) optimistic start

- In 1966, Prof. Marvin Minsky at MIT asked his undergraduate student Gerald Jay Sussman to: spend 3 months in the summer to link a camera to a computer & get the computer to ‘describe what it saw’

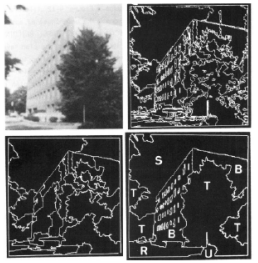
- In ~2026 (~60y later), we know that this problem is *significantly* more challenging

Brief History

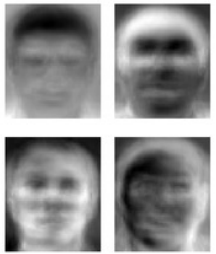
- **1963:** Robert's thesis
- **1966:** Minsky assigns computer vision as an undergrad summer project
- **1960s:** Interpretation of synthetic worlds
- **1970s:** Some progress on interpreting selected images
- **1980s:** ANNs come and go; shift toward geometry and increased mathematical rigor, Marrs book published (after he died).
- **1990s:** Face recognition; Statistical analysis
- **2000s:** Broader recognition; Large annotated datasets available; video processing starts; Vision & graphics; HCI; Internet vision, etc
- **2012:** AlexNet, Kinect, big-data, Google Car, …
- **2013~:** Deep learning, Image-Net, …
- **2020~:** Transformer, Vision+Language, Diffusion model …



Guzman '68

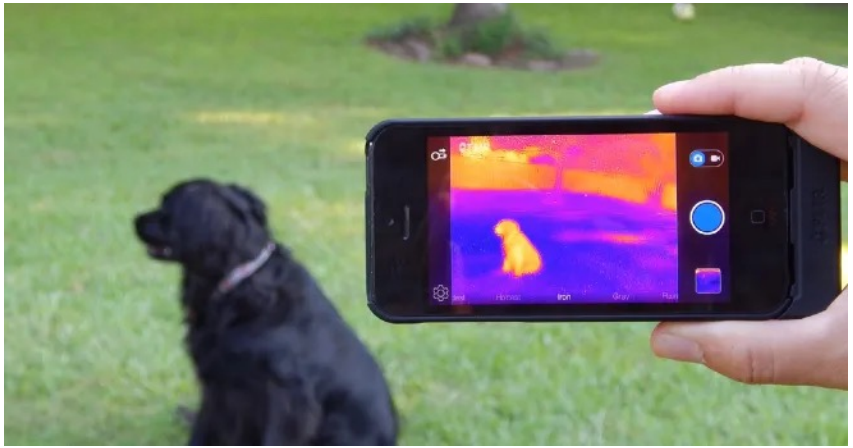


Ohta Kanade '78

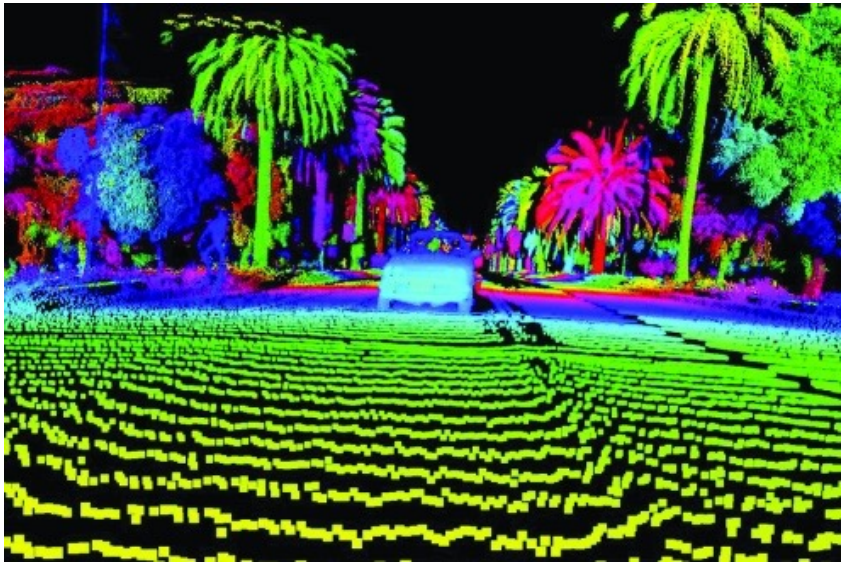


Turk and Pentland '91

Many other vision sensors



Infra Red



Lidar

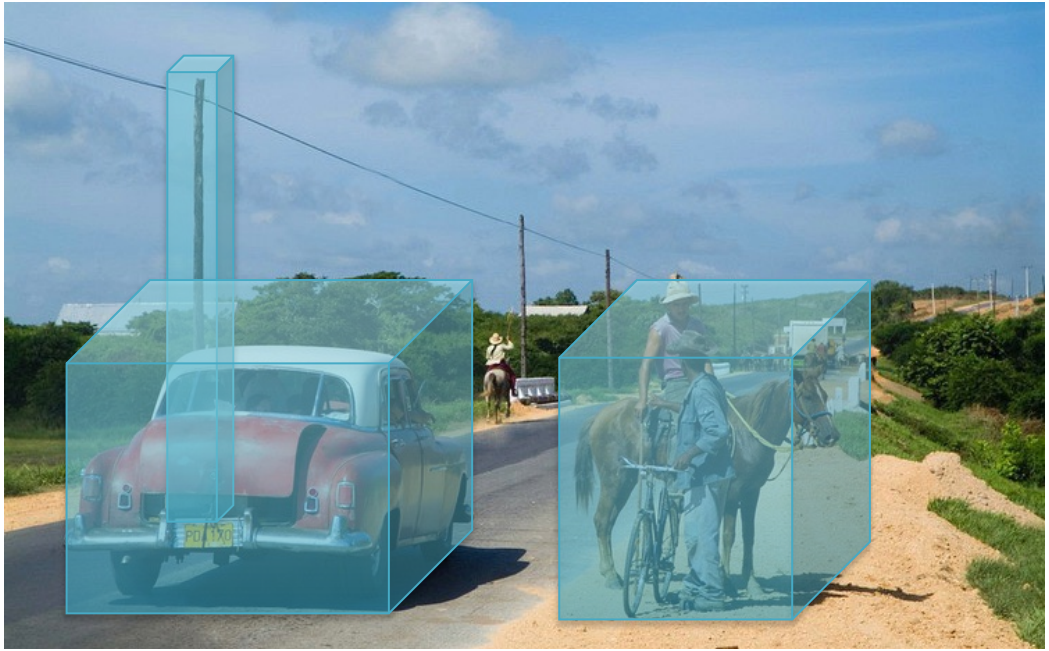
Computer Vision

- **Low-Level Vision**
  - Color
  - Contrast
  - Edges
- Mid-Level Vision
  - 3D reconstruction
  - Depth
  - Motion
- High-Level Vision
  - Object classification
  - Activity recognition
  - Instance segmentation



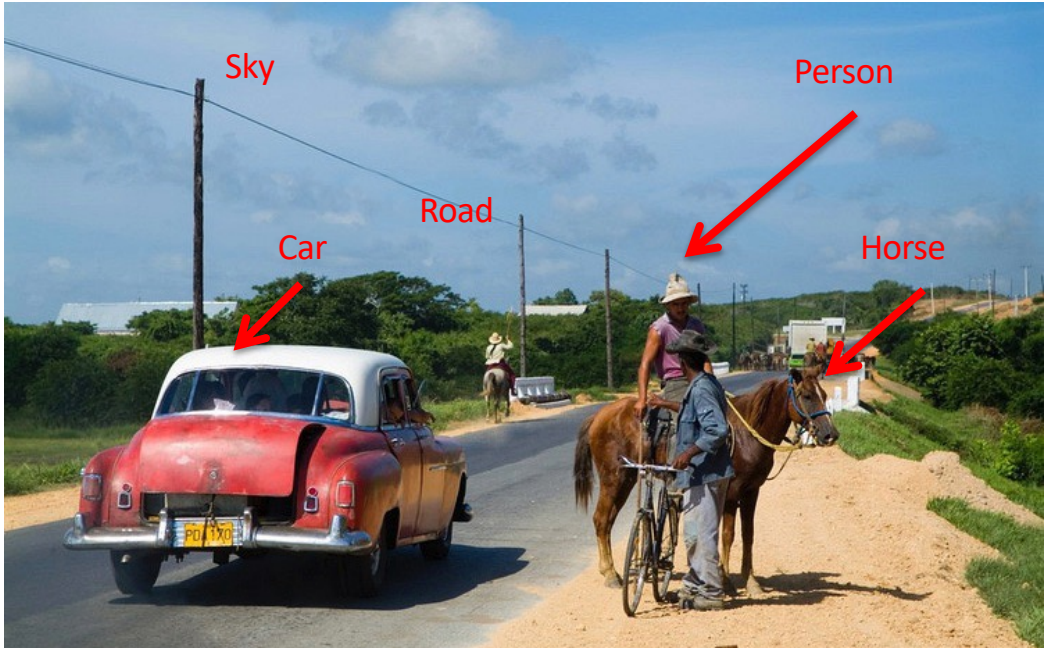
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Computer Vision

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Applications

Panorama Stitching

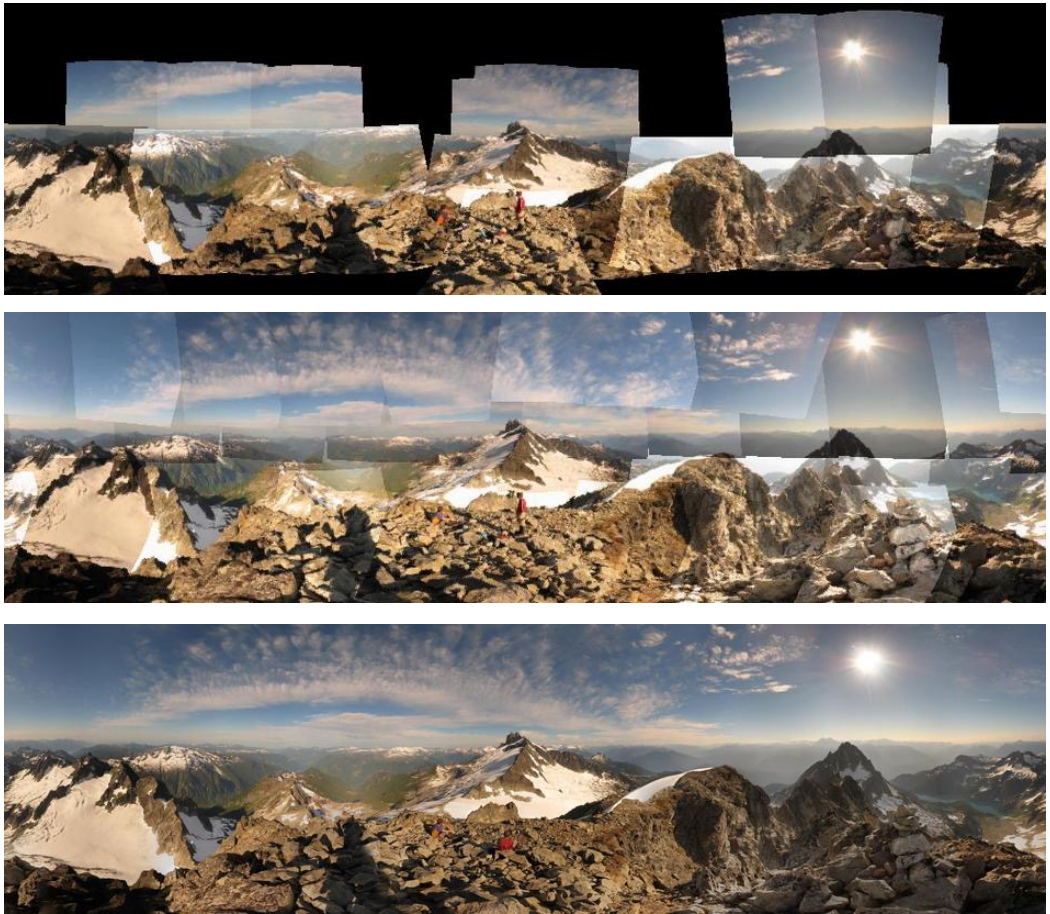


Image Segmentation



SAM
https://segment-anything.com
[Kirillov et al., ICCV'23]

Image Segmentation – Language + Vision



[Kerr et al., LERF: Language Embedded Radiance Fields, ICCV'23]

Image Inpainting – ‘Magic Eraser’



https://advimman.github.io/lama-project

[Suvorov et al., WACV’22]

Point Tracking



https://omnimotion.github.io

[Wang et al., ICCV'23]

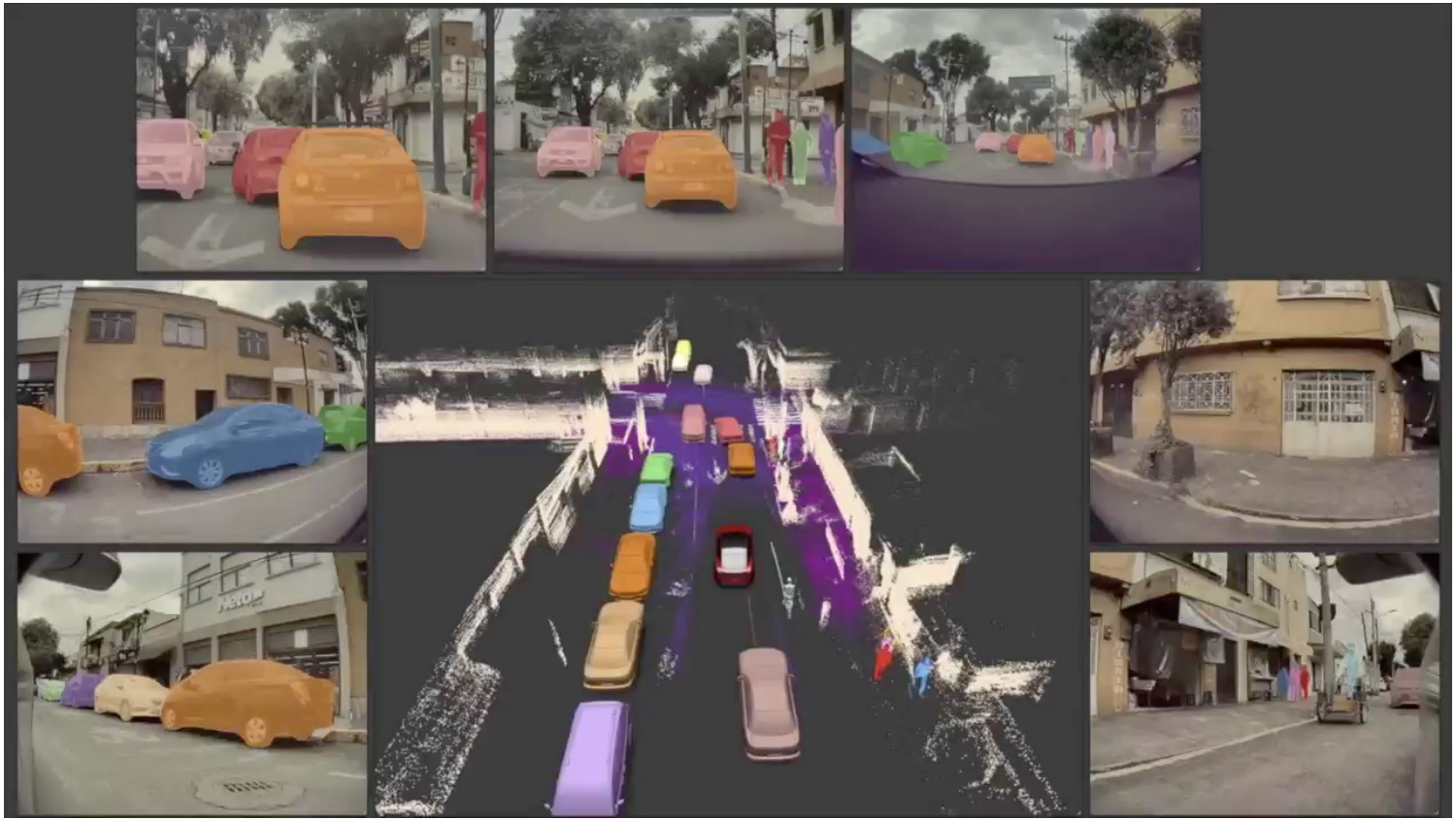
Depth Estimation from Color Image



https://marigoldmonodepth.github.io

[Ke et al., CVPR'24]

Autonomous Driving



[Tesla AI, 2022]

3D Humans from a Single Image



Expressive Body Capture: 3D Hands, Face, and Body from a Single Image

G. Pavlakos* · V. Choutas* · N. Ghorbani · T. Bolkart · A. A. A. Osman · D. Tzionas · M. J. Black

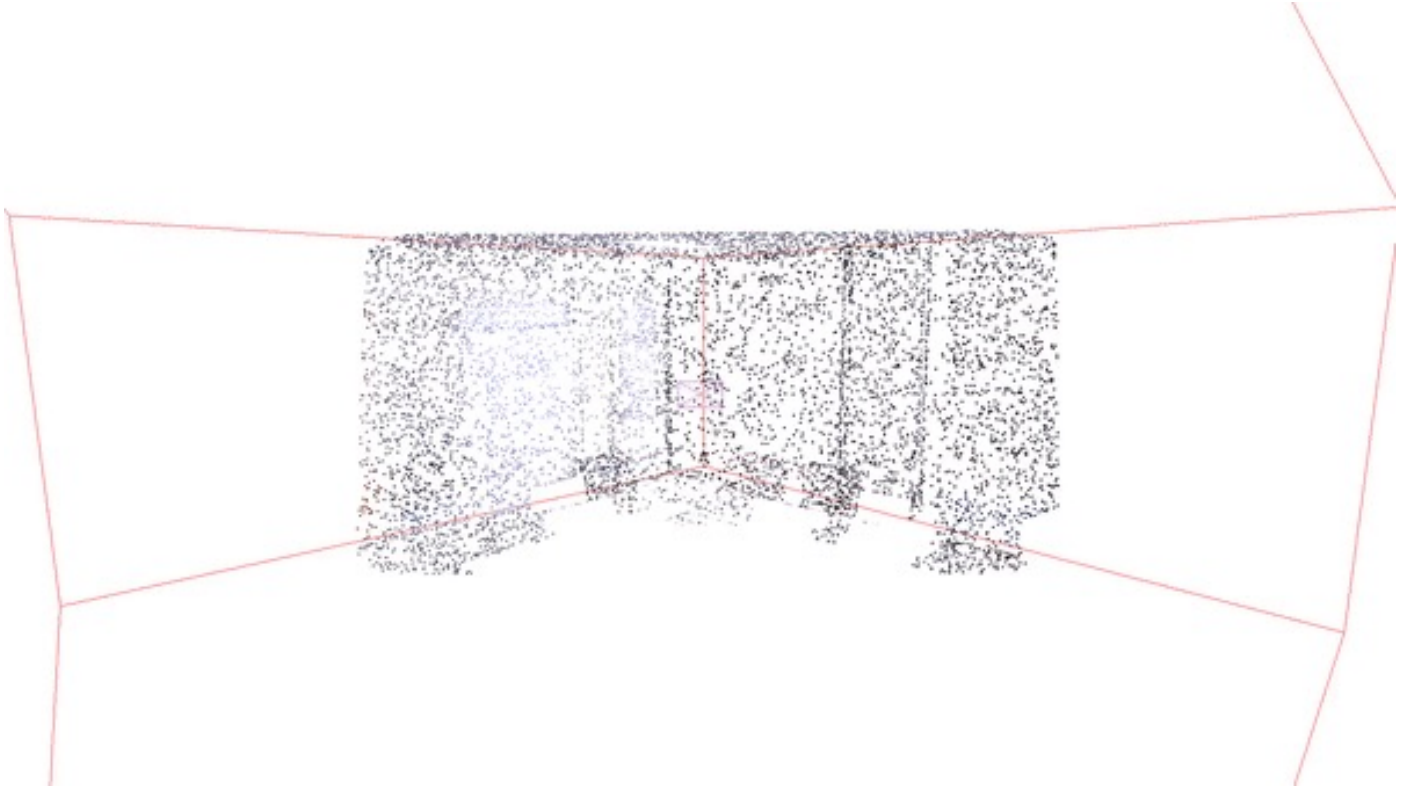
CVPR 2019

3D Humans from a Single Video



Resolving 3D Human Pose Ambiguities with 3D Scene Constraints
Mohammed Hassan, V. Choutas, D. Tzionas and M. J. Black
ICCV 2019

3D Space Reconstruction (SLAM)

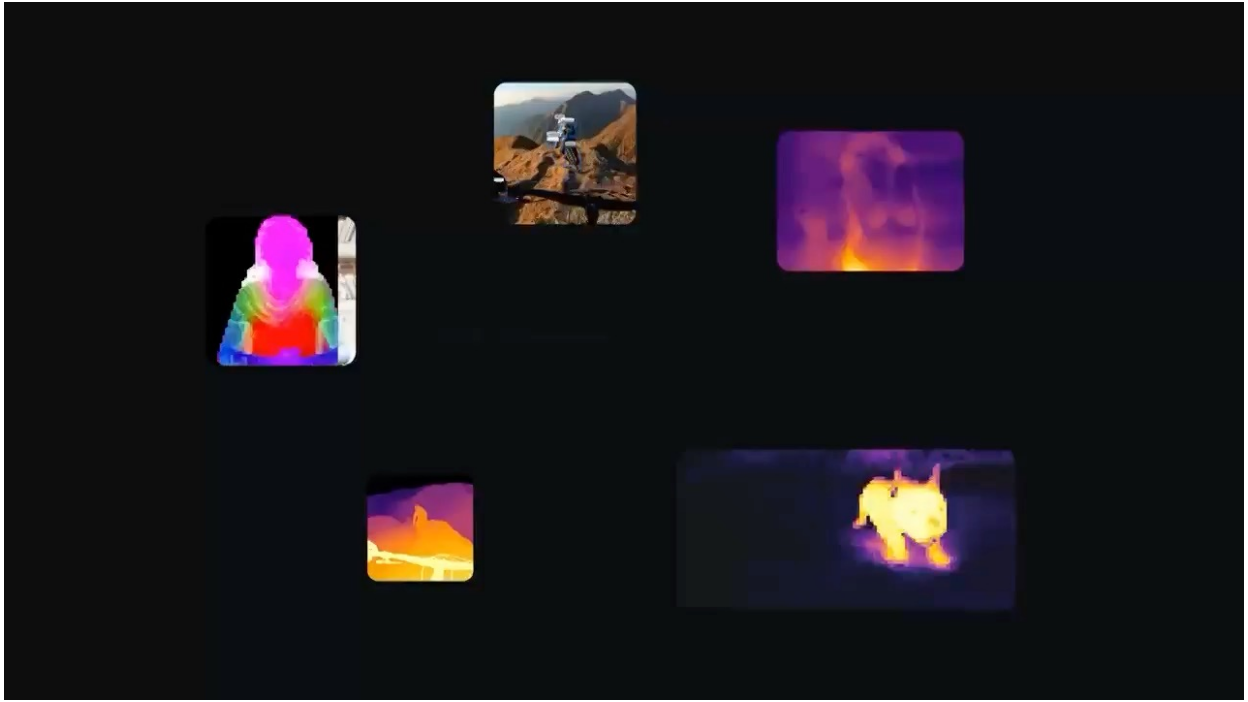


Point-SLAM: Dense Neural Point Cloud-based SLAM

Erik Sandström* · Yue Li* · Luc Van Gool · Martin R. Oswald

ICCV 2023

Foundational Models

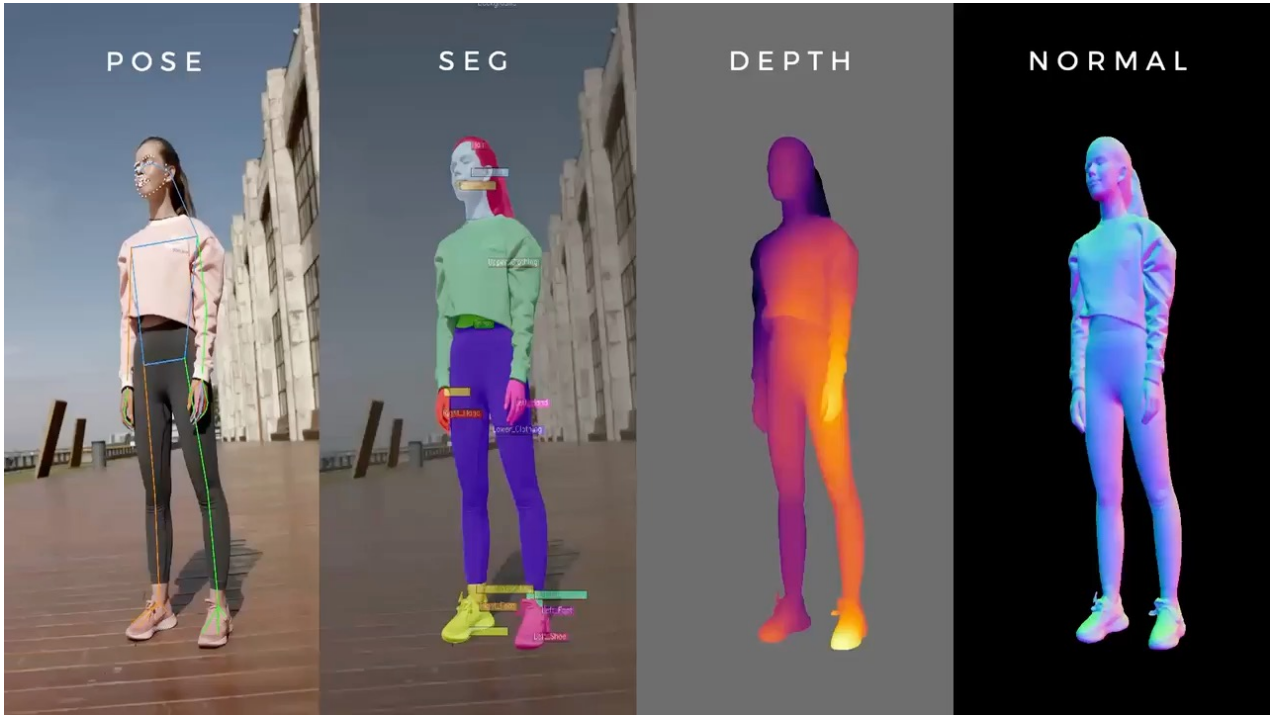


DINOv3

https://github.com/facebookresearch/dinov3

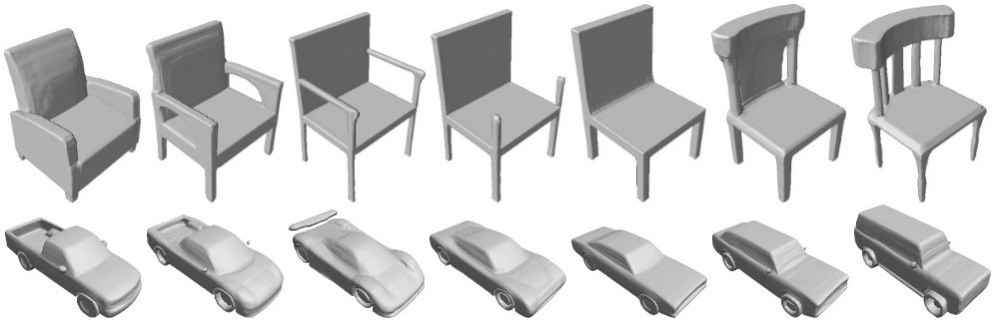
[Siméoni et al., arXiv:2508.10104, 2025]

Foundational Models → Humans

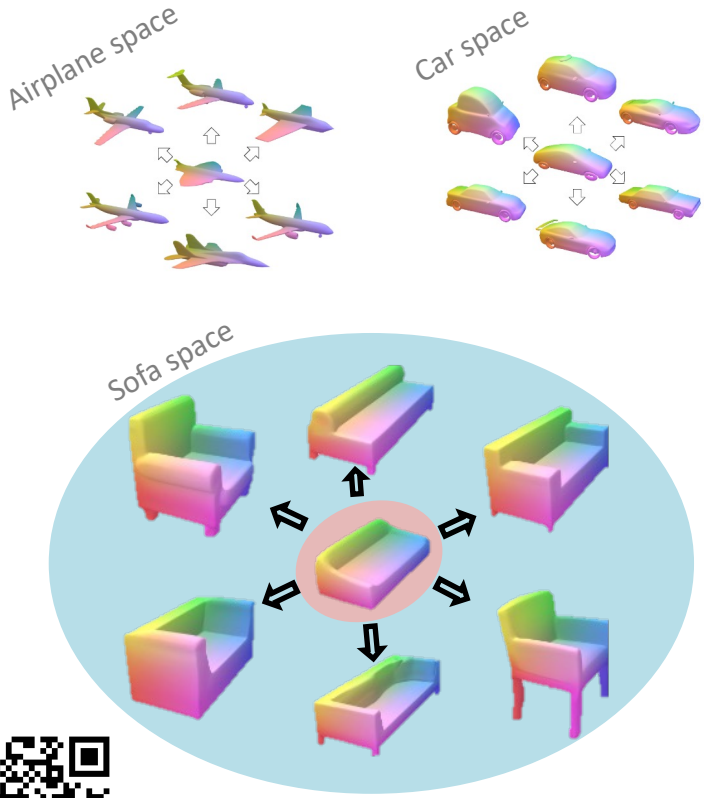


SAPIENS: Foundation for Human Vision Models
https://github.com/facebookresearch/sapiens
[Khirodkar et al., ECCV 2024]

3D Object Shape Modelling

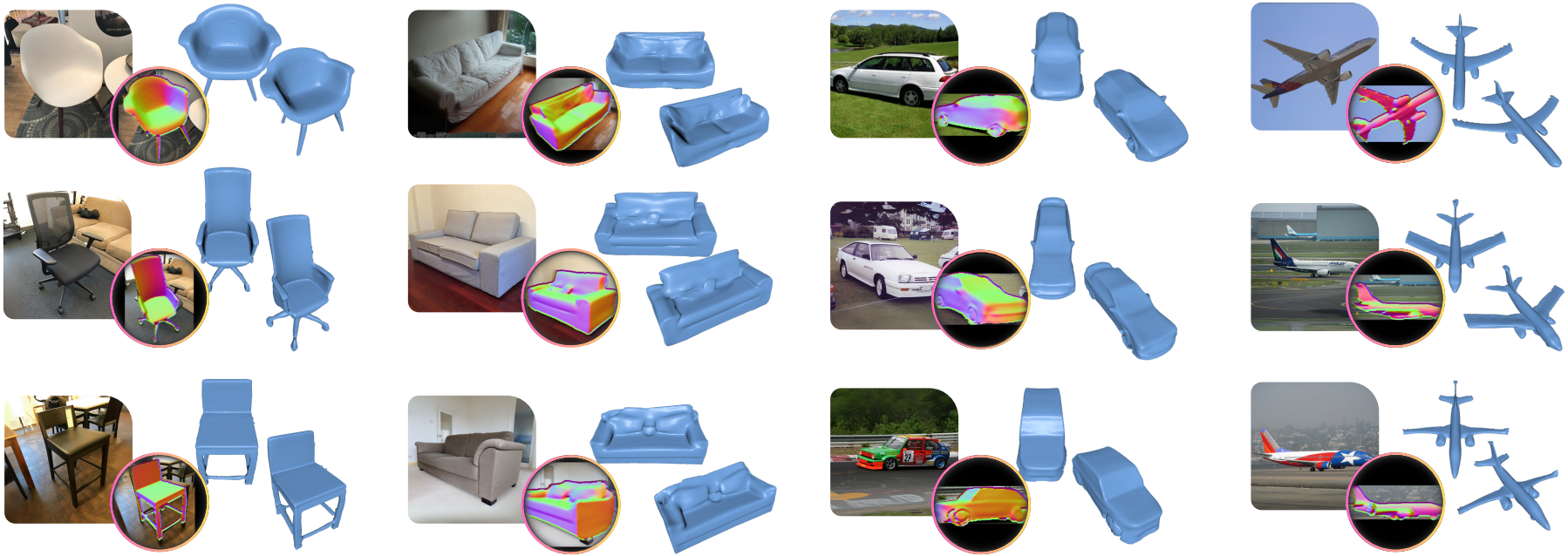


DeepSDF
[J.J.Park et al., CVPR]



Deep Implicit Templates (DIT)
[Zheng et al., CVPR 2021]

3D Object from a Single Image



SDFit: 3D Object Shape and Pose by Fitting a Morphable SDF to a Single Image

Dimitrije Antić · G. Paschalidis · S. Tripathi · T. Gevers · S. K. Dwivedi · D. Tzionas

ICCV 2025

Reenactment – aka DeepFakes



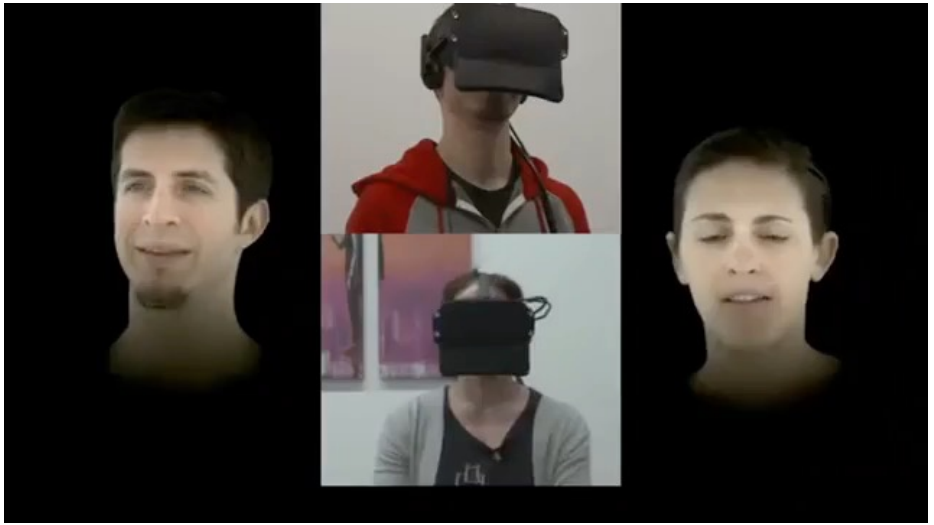
Face2Face [Thies et al., CVPR 2016]

Mixed Reality: 3D Video Conferencing



Microsoft HoloPortation, 2016

[https://www.microsoft.com/en-us/research/project/holoportation-3]



Facebook Reality Labs - Lombardi et al., TOG 2018

[https://www.youtube.com/watch?v=gpdX9jkhv2U]

Image-based 3D Reconstruction



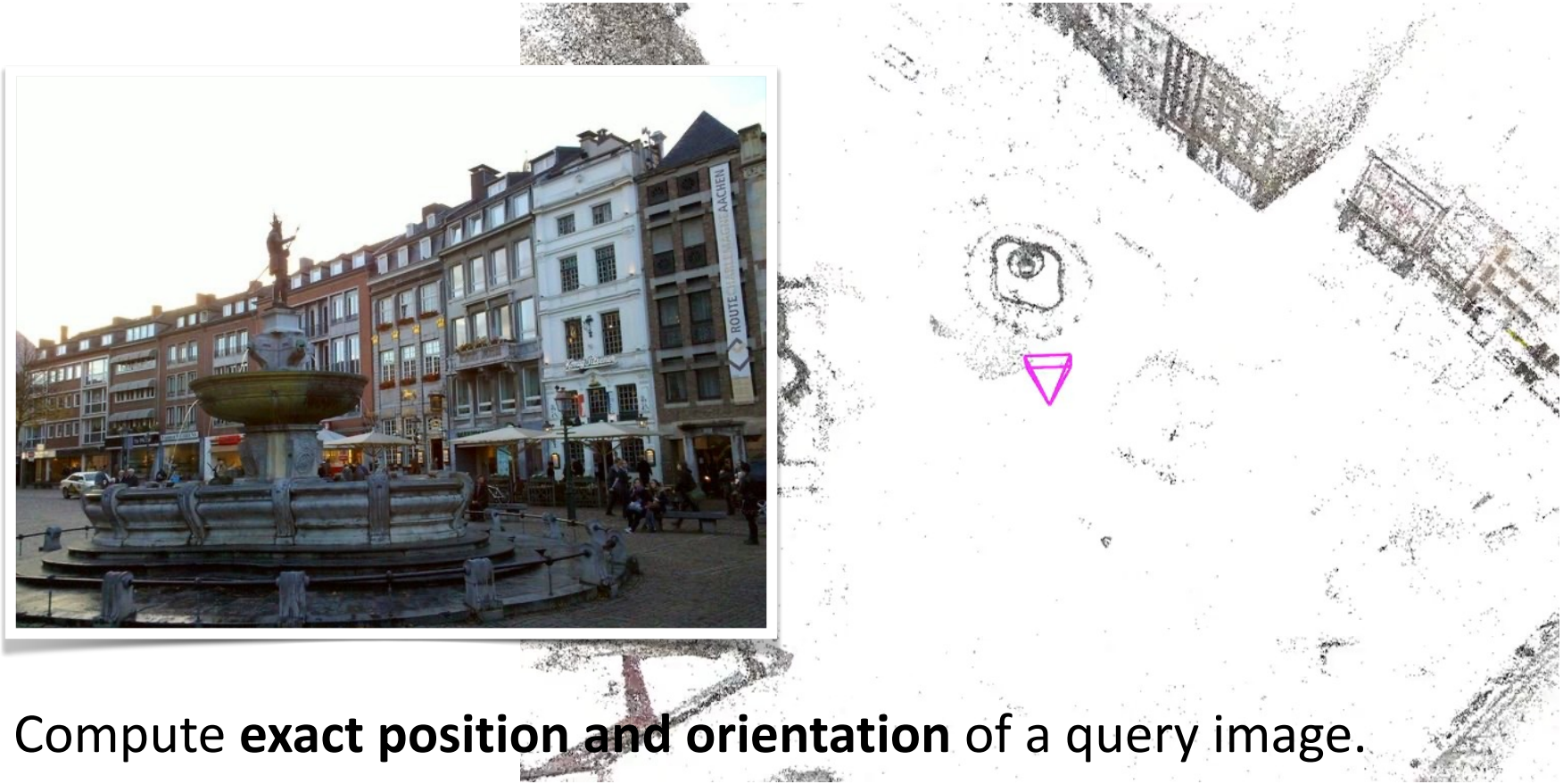
Building Rome in a Day

S. Agarwal, N. Snavely, I. Simon, S. M. Seitz, R. Szeliski

ICCV 2009



Image-based Localization



Compute **exact position and orientation** of a query image.

Generative Models (Text → Video) – Success



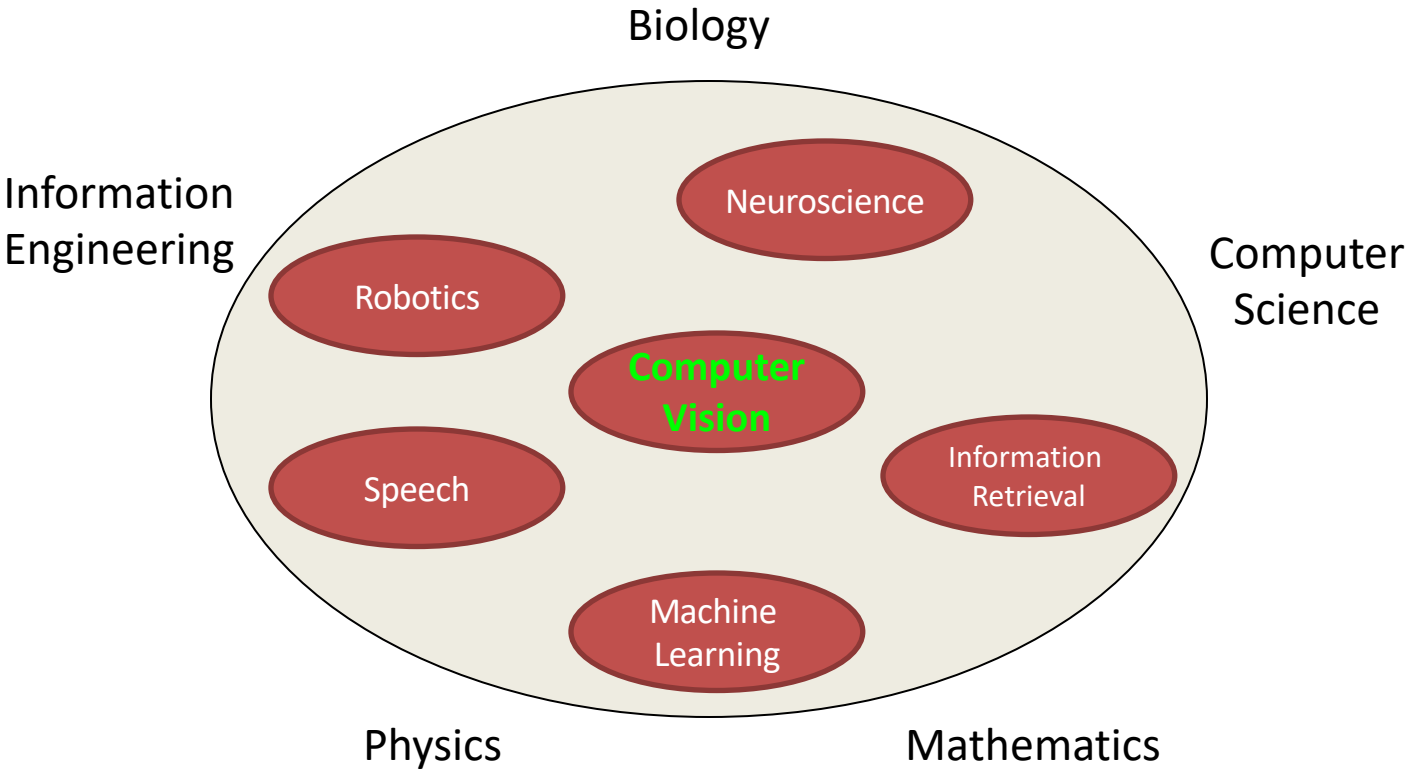
[OpenAI, SORA] [Examples, 2024]

Generative Models (Text → Video) – Failure



[OpenAI, SORA] [Examples, 2024]

Multi-disciplinary Research Field



Course Info

Course Team

Teachers:

- Fivos Tzavellos (manages course administration tasks and TAs)
- Yen-Chia Hsu (coordinate the course management and gives lectures)
- Arnoud Visser (gives lectures)
- Arun Mukundan (gives seminars)

Teaching Assistants (handle Labs)

- Julia Belloni
- Ardjano Mark
- Kaan Sert
- Lisanne Wallaard
- Kristian Johansen
- Ippokratis Pantelidis
- Mattia Sferrazza
- Lev Hulsbergen

Schedule Outline

Week	Monday	Tuesday	Wednesday	Thursday	Friday
36 31/8 4/9		9-10:45 HC: Lecture 1 Y. Hsu SP L1.02		13-14:45 HC: Lecture 2 Y. Hsu SP L1.02 15-16:45 Computerpracticum 1, 2: SP D1.111	9-10:45 Computerpracticum 7, 8: SP C0.110 13-14:45 Computerpracticum 3, 4: SP F2.04 15-16:45 Computerpracticum 5, 6: SP G1.18
37 7/9 11/9		9-10:45 HC: Lecture 3 Y. Hsu SP L1.02	15-16:45 WC: Seminar 1 A. Mukundan SP L1.02	11-12:45 HC: Lecture 4 Y. Hsu SP C1.110 13-14:45 Computerpracticum 1, 2: SP L1.02	9-10:45 Computerpracticum 7, 8: SP C0.110 13-14:45 Computerpracticum 3, 4: SP G1.18 15-16:45 Computerpracticum 5, 6: SP G1.18
38 14/9 18/9		13-14:45 HC: Lecture 5 dr. Arnoud Visser SP L1.02	11-12:45 WC: Seminar 2 A. Mukundan SP C0.05	11-12:45 HC: Lecture 6 dr. Arnoud Visser SP C1.110 13-14:45 Computerpracticum 1, 2: SP L1.02	9-10:45 Computerpracticum 7, 8: SP C0.110 13-14:45 Computerpracticum 3, 4: SP G1.18 15-16:45 Computerpracticum 5, 6: SP G1.18
39 21/9 25/9		13-14:45 HC: Lecture 7 Y. Hsu SP C1.110	15-16:45 WC: Seminar 3 A. Mukundan SP L1.02	11-12:45 HC: Lecture 8 Y. Hsu SP C1.110 13-14:45 Computerpracticum 1, 2: SP L1.02	9-10:45 Computerpracticum 7, 8: SP C0.110 13-14:45 Computerpracticum 3, 4: SP G1.18 15-16:45 Computerpracticum 5, 6: SP A1.16
40 28/9 2/10		9-10:45 HC: Lecture 9 dr. Arnoud Visser SP L1.02	15-16:45 WC: Seminar 4 A. Mukundan SP L1.02	11-12:45 HC: Lecture 10 dr. Arnoud Visser SP C1.110 13-14:45 Computerpracticum 1, 2: SP L1.02	9-10:45 Computerpracticum 7, 8: SP C0.110 13-14:45 Computerpracticum 3, 4: SP G1.18 15-16:45 Computerpracticum 5, 6: SP A1.16
41 5/10 9/10		9-10:45 HC: Lecture 11 dr. Arnoud Visser SP L1.02	11-12:45 WC: Seminar 5 A. Mukundan SP H0.08	11-12:45 HC: Lecture 12 dr. Arnoud Visser SP C1.110 13-14:45 Computerpracticum 1, 2: SP A1.16	9-10:45 Computerpracticum 7, 8: SP C0.110 13-14:45 Computerpracticum 3, 4: SP A1.16 15-16:45 Computerpracticum 5, 6: SP A1.16
42 12/10 16/10		9-10:45 HC: Lecture 13 dr. Arnoud Visser SP L1.02	15-16:45 WC: Seminar 6 A. Mukundan SP L1.02		9-12 Tentamen digitaal Y. Hsu IWO 4.04B (Geel)

Administration

- Announcements will be made on Canvas.
- Lectures will be recorded with links on the Canvas home page (quality not guaranteed).
- Lectures may be given virtually if unexpected situations happen, same as seminars.
- Use Ed Discussion (preferred) or email to ask questions.
- There is no attendance requirement.
- You are expected to treat others with mutual respect and appreciation regardless of any differences.
- It is strongly recommended to stay home if you have symptoms associated with respiratory infections.

Learning Approach

- All materials will be on Canvas
- HC – Lectures
  - Cover fundamental Computer Vision concepts, Math foundations, Applications
  - Focus on higher-level concepts and intuitions
  - MSc → Lectures will not cover everything → self study (slides, textbook, search online) is important!
- WC – Seminars
  - Practice theoretical problems
  - Final week – discuss mock exam
- Lab Assignments (not graded)
  - Design and Implement a hands-on solution for fundamental Computer Vision problem
- Computerpracticum – Computer Labs
  - Receive help for Lab Assignment → Free-form structure

Intended Learning Outcomes – Lectures

- Understand the basic concepts and motivation of Computer Vision
- Understand the key theories & techniques
- Know the *trends* of modern Computer Vision
- Gain the basis knowledge for *future* courses, such as:
  - Computer Vision 2
  - Deep Learning

Intended Learning Outcomes – Seminars

- Use key theories & techniques with emphasis on *theory*
- Reflect on the pros and cons of algorithms
- Get prepared for future courses

Intended Learning Outcomes – Labs

- Use key theories & techniques with emphasis on *practice*
- Propose improvements over existing methods
- Get prepared for future courses

Lecture Schedule

Week	Date		Lecturer
1	1 Sep (Tue)	Lecture 1: Introduction & Course Info	Yen-Chia
	3 Sep (Thu)	Lecture 2: Camera, Image formation, and Color	Yen-Chia
2	7 Sep (Tue)	Lecture 3: Reflection, Photometric Stereo, and Image Processing	Yen-Chia
	10 Sep (Thu)	Lecture 4: Neighborhood Processing and Image Filtering	Yen-Chia
3	15 Sep (Tue)	Lecture 5: Local Features, Edges, Lines, and Corners	Arnoud
	17 Sep (Thu)	Lecture 6: Optical Flow, Motion and Tracking	Arnoud
4	22 Sep (Tue)	Lecture 7: Global Transformations and Image Stitching	Yen-Chia
	24 Sep (Thu)	Lecture 8: Image Classification	Yen-Chia
5	29 Sep (Tue)	Lecture 9: Retrieval, Detection and Segmentation	Arnoud
	1 Oct (Thu)	Lecture 10: Convolution Neural Networks, Object Detection Basics	Arnoud
6	6 Oct (Tue)	Lecture 11: Single Shot Detection, Network Architectures	Arnoud
	8 Oct (Thu)	Lecture 12: Shape from X, Stereo, Structure from Motion	Arnoud
7	13 Oct (Tue)	Lecture 13: Guest Lecture and Exam Q&A	Arnoud
	16 Oct (Fri)	Final Exam	Yen-Chia

Grading

- Assessment is based on one final exam (100%).
- You need to get at least 5.5 (out of 10) to pass the course.
- Check the syllabus for details (e.g., grade calculation).

Exam

- All lecture slides, seminar materials (exercises), and labs will be materials for the exam (except the guest lecture). We provide mock exams for you to practice (but they are not representative of this year's exam because of changes in grading and the exam method).
- Exams are based on Multiple choice question (select one answer from several options), Multiple response question (select one or multiple answers from several options), and Numerical value question (provide a numerical value as the answer).
- You may bring THREE A4-size cheat sheet with two sides of content (written or printed) to the exams. No other materials are allowed (except the cheat sheet). Also, no calculators.
- Check the date and time of the exams carefully on UvA DataNose. There is a resit, which will override your original exam score.
- We will use guess correction for the exams (check the syllabus for details). You will have access to Jupyter Notebook during the exam.
- Check the syllabus for details.

GenAI Usage and Policy

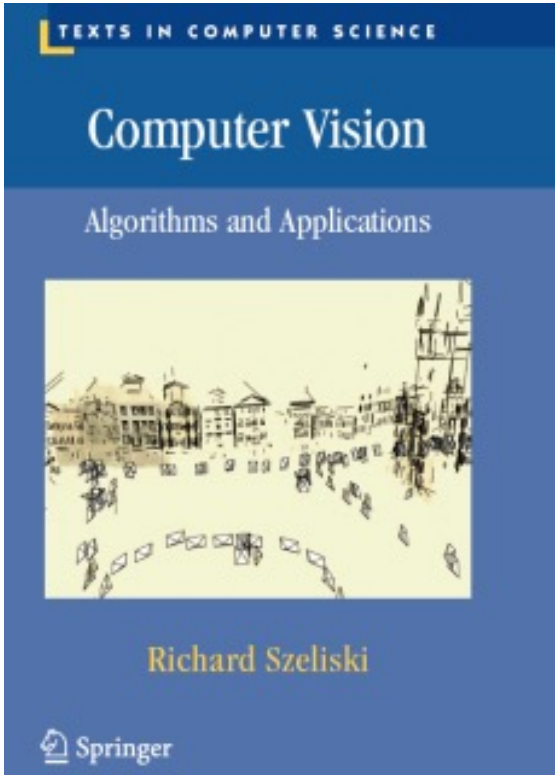
- We allow students to use GenAI tools, such as ChatGPT, Gemini, Claude, UvA AI Chat, etc.
- GenAI tools can hallucinate information, generate wrong content, or provide fake sources/citations. You are responsible for fact-checking and verifying all AI-generated content.
- The official lecture slides and course readings remain the single source of truth. If the AI generates content that contradicts the course materials, you must follow the course materials.
- The teaching team is neither responsible nor liable for your exam errors resulting from reliance on unverified AI-generated content.
- Check the syllabus for details.

Math, Coding, and Course Readings in the Exam

- If a math equation is on the slide, and we explained it in the lectures, seminars, or labs, it means that the equation may appear in the exam (in different ways).
- Questions in the exam can contain coding-related questions.
- This course has many readings. Materials in the readings may also appear in the exam. The exams will be mainly based on the lecture slides, seminar materials (exercises), and labs.
- Check the Schedule outline on the course syllabus for the required readings.
- If a concept appears on the lecture slides, seminar materials (exercises), **or** labs, it may appear on the exam, and the course readings are used to support the concept with more explanations.
- If a concept appears in the course readings but not on the slides (also not in seminar materials and not in the labs), it will not appear on the exam.
- Check the syllabus for details.

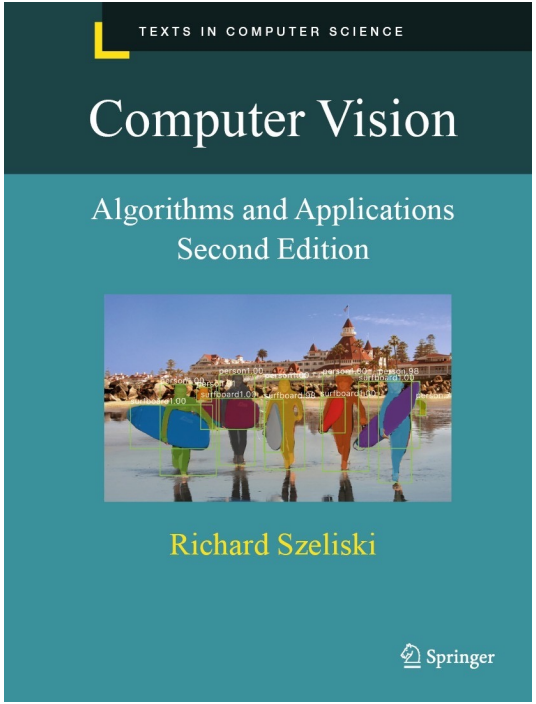
Textbook

- Computer Vision:
  Algorithms and Applications
  - by Richard Szeliski
  - freely available to download at http://szeliski.org/Book



Reading for this lecture

- Sec. 1.1
- Sec. 1.2



Communication

- When you have questions regarding administration:
  - → Step 1: Check syllabus first for answers
  - If Step 1 does not work → Step 2: Contact your TA (every study group has a TA) or ask the question on Ed Discission on Canvas
  - If Step 2 does not work → Step 3: Contact the senior TA (Fivos Tzavellos, email on Canvas home page).
  - If Step 3 does not work → Step 4: Contact the course coordinator (Yen-Chia Hsu, email on Canvas home page)
- Please kindly follow the communication chain and avoid contacting the course coordinator directly.

Communication

- When you have questions regarding the course content:
  - → Step 1: Check the course readings for more information, or check other available online educational materials (e.g., courses from other institutions)
  - If Step 1 does not work → Step 2: ask the question on Ed Discission on Canvas, and the teaching team will answer it. You can also ask the teachers or TAs directly during the lectures, seminars, or computer labs.

Closing Remarks

Action Points

- Check information regularly on Canvas
- Canvas announcements – They overwrite everything!
- Download the textbook
- Install Python Jupyter Notebook & practice NumPy & OpenCV
- Check the syllabus before asking questions

Disclaimer

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Questions?